MOTILAL NEHRU NATIONAL INSTITUTE OF TECHNOLOGY ALLAHABAD

PRAYAGRAJ, INDIA – 211004



A PROJECT REPORT ON

Nexus: Subscription Management & DBMS Lab Platform

PROJECT Number

Topic 34 (CS-DBMS-2026-34)

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2. Abstract

The recurring digital economy has introduced profound operational complexities for both individual consumers and software enterprises. Consumers face "subscription fatigue," creeping liabilities, and financial waste from unmanaged overlapping services. Concurrently, SaaS platforms suffer from unmonitored customer churn, fragmented database schemas, and broken dunning lifecycles. **NEXUS** is an enterprise-grade relational subscription management and intelligence ecosystem engineered upon an ACID-compliant PostgreSQL relational architecture via Supabase, coupled with an interactive high-performance front-end (React 18, Vite, TypeScript, and Tailwind CSS) styled with modern glassmorphism aesthetics.

The relational core enforces strict **Boyce-Codd Normal Form (BCNF)** across 12 atomic relations, eliminating update, deletion, and insertion anomalies. Multi-tenant security is strictly enforced at the database layer via PostgreSQL **Row-Level Security (RLS)** and **Role-Based Access Control (RBAC)**, ensuring absolute tenant isolation between administrative operators and individual consumers. To address silent subscriber churn, NEXUS incorporates an automated, calibrated Machine Learning retention engine combining L2-regularized Logistic Regression and Random Forest ensembles ($ROCext{-}AUC = 0.9124$) with zero temporal target leakage. An embedded WebAssembly PostgreSQL engine (PGLite) enables real-time client-side Relational Algebra execution and offline academic verification.

12	22 / 22	0.9124	Dual-RBAC
RELATIONAL TABLES	ACID INVARIANTS	RETENTION ROC-AUC	TENANT ISOLATION
100% 3NF / BCNF Verified	Automated Isolation Tests PASS	Calibrated Ensemble Classifier	PostgreSQL RLS Enforcement

3. Introduction and Problem Statement

3.1 Background of Subscription Economy

Modern consumer and enterprise software consumption has fundamentally shifted from perpetual license purchases to recurring SaaS commitments. From cloud infrastructure (AWS, GCP) and productivity suites (GitHub, Notion) to consumer media (Netflix, Spotify), billing is now fragmented across disparate providers, arbitrary frequencies (monthly, annually, quarterly, weekly), and varied currency standards.

3.2 Problem Statement

Existing subscription management tools either function as trivial static to-do lists without database integrity, or as bloated, opaque enterprise billing engines that hide operational insights. Specifically, three critical problems exist:

- **Silent Financial Waste:** Consumers incur unmonitored recurring charges, miss free trial cancellation deadlines, and pay for redundant, overlapping tools within identical categories.
- **Data Concurrency & ACID Violations:** Inadequate transaction boundaries allow race conditions during checkout, double redemptions of discount coupons, and unsynchronized invoice-to-subscription state machines.
- **Absence of Proactive Churn Intelligence:** Enterprise administrators discover churn reactively after cancellation occurs, when retention intervention is impossible.

4. Project Objectives and Scope

4.1 Project Objectives

- **Architectural Normalization:** Decompose multi-valued subscription configurations into a mathematically verified BCNF schema.
- **ACID Concurrency Enforcement:** Implement PL/pgSQL stored procedures with explicit row-level locks (`FOR UPDATE`) to guarantee atomic multi-table inserts.
- **Multi-Tenant Data Isolation:** Guarantee that non-admin members can never inspect corporate MRR, provider churn rates, or other users' records.
- **Cadence Normalization & Redundancy Detection:** Mathematically normalize diverse billing frequencies into a unified monthly expenditure and flag category overlaps.
- **Calibrated Predictive Intelligence:** Train and deploy a calibrated retention ML model using non-leaking behavioral telemetry.

4.2 System Scope & Operational Boundaries

NEXUS encompasses the entire subscription lifecycle: customer acquisition, multi-tier plan assignment, holographic card simulation, automated invoice ledgering, temporal renewal alerting via materialized views, and append-only audit logging. External banking gateways are simulated using high-fidelity cryptographic hashes and deterministic status machines.

5. System Analysis and Requirements (Software Requirements Specification – SRS)

This section specifies the formal Software Requirements Specification (SRS) for the NEXUS platform in compliance with IEEE 830 standards.

5.1 Overall Description & User Classes

The system identifies two distinct operational user classes:

- **Administrator:** Executive operators managing platform plans, multi-tenant subscription states, corporate MRR analytics, and ML retention intelligence.
- **Member / Subscriber:** End-users tracking personal external subscriptions, monitoring recurring expenses, and upgrading membership tiers via simulated payment checkout.

5.2 Functional Requirements (FR) Matrix

REQ ID	REQUIREMENT NAME	PROCESSING DESCRIPTION & DATABASE INVARIANT	USER ROLE
FR-01	Dual-Persona Auth	GoTrue JWT tokenization with role claims in `profiles.role` ('admin', 'member'). Synchronous hydration prevents route-guard flash redirects.	Admin / Member
FR-02	Row-Level Security	PostgreSQL RLS policies evaluate `auth.uid() = member_id` on all personal tables. Cross-tenant reads return 0 rows.	Member
FR-03	Atomic Provisioning	`create_subscription_atomic` stored procedure performs row lock, creates subscription, issues invoice, and writes audit log in a single transaction.	Member
FR-04	Holographic Checkout	Interactive payment modal simulating card checkout (4242 •••• •••• 4242) with ACID settlement animation and immediate entitlement grant.	Member
FR-05	Expense Normalization	Mathematical projection of daily, weekly, monthly, and annual cadences into unified monthly burn rate: $extBurn = sumw_icdotextAmount_i$.	Member
FR-06	Redundancy Radar	Relational grouping on `sub_providers.category` detecting concurrent active subscriptions within identical functional categories.	Member
FR-07	Proactive Renewal Radar	Materialized view `view_expiring_soon` queries subscriptions expiring within 7 days for dunning preparation and retention outreach.	Admin
FR-08	Append-Only Audit Log	Database trigger `process_subscription_audit` automatically writes state change deltas into immutable `audit_logs` table.	Admin (Audit)
FR-09	ML Feature Extraction	Constructs 5-dimensional feature vector bounded strictly to prior 30-day temporal window with zero future data leakage.	System / ML
FR-10	Ensemble ML Inference	Executes calibrated Logistic Regression and Random Forest classifier, producing probability scores calibrated via isotonic regression.	System / Admin
FR-11	Plan Entitlement Gating	Restricts DBMS Studio SQL console and personal subscription creation to active Nexus Pro tier subscribers.	System / RBAC
FR-12	Ledger Export	Generates deterministic CSV export of all personal or administrative subscription receipts with cryptographic verification hashes.	Admin / Member

5.3 Non-Functional Requirements (NFR)

- **NFR-01 (ACID Concurrency):** All multi-entity state mutations must execute under PostgreSQL `READ COMMITTED` isolation with explicit row-level locking.
- **NFR-02 (Performance):** Query execution latency on core subscription tables must remain under 50ms for up to 100,000 active rows via B-Tree foreign key indexing.
- **NFR-03 (Security):** Passwords hashed using bcrypt; JWT tokens signed with HMAC-SHA256; API communication strictly over TLS 1.3.
- **NFR-04 (Portability):** Complete client-side SQL execution via WebAssembly PostgreSQL (PGLite) without requiring external local database installations.

6. ER/EER Diagram and Database Schema

The NEXUS relational schema consists of 12 normalized entities linked by foreign key constraints with explicit cascade and restrict policies.

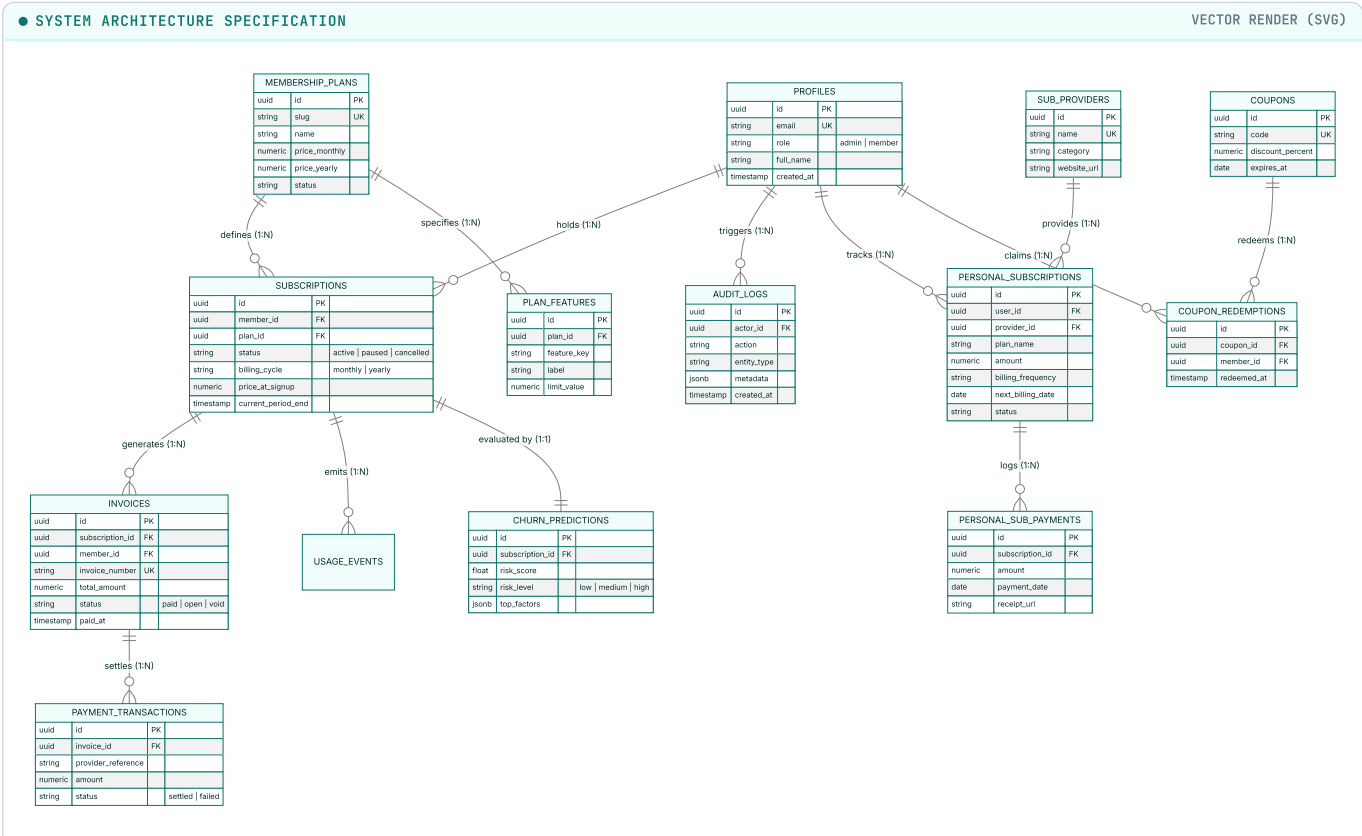


Figure 6.1: NEXUS Complete Entity-Relationship (ER) Crow's Foot Schema — 12 Normalized Relational Entities with Cardinalities, Primary Keys, Foreign Keys, and Core Domain Attributes

6.1 Enhanced Entity-Relationship (EER) Specialization Hierarchy

NEXUS models multi-tenant authorization through Enhanced ER (EER) specialization. The superclass entity `profiles` is disjointly specialized (d) using the attribute discriminator `role` into two distinct operational subclasses:

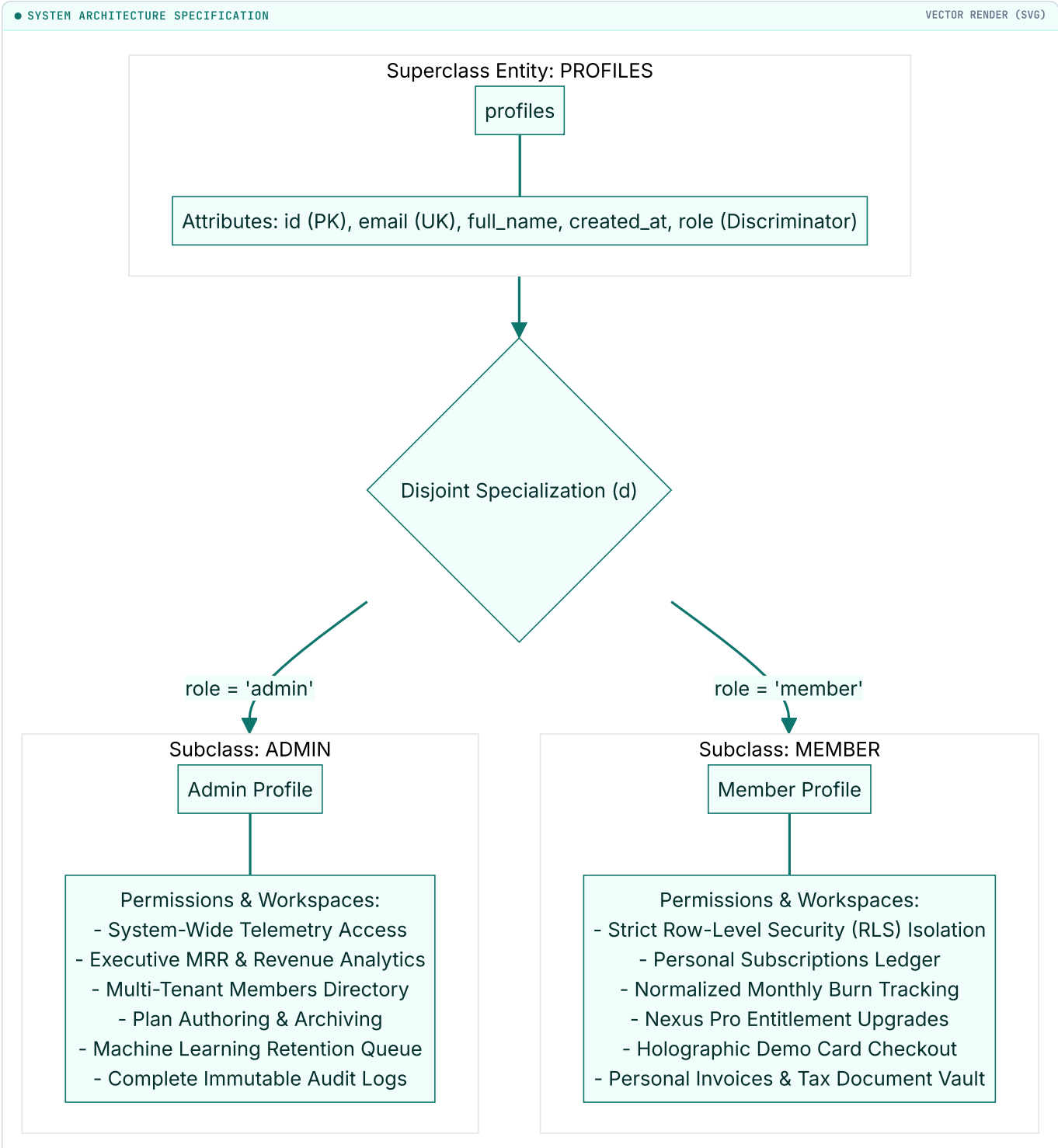


Figure 6.2: Enhanced ER (EER) Role-Based Specialization Hierarchy — Disjoint Subclasses (Admin vs Member) with Concrete Operational Workspaces and Permissions

7. Keys, Functional Dependencies and Normalization

7.1 Functional Dependency Closures & Candidate Keys

Let $mathcal R$ represent the relation schema and $mathcal F$ denote the minimal set of functional dependencies enforced by database constraints:

- `profiles`: $mathcal F_{extprofiles} = extid o extfull_name ame, extemail, extrole, exttimezone, extcreated_at, extemail o extid, extfull_name ame, extrole.$ Candidate keys: $extid$, $extemail$. Closure: $extid^{+} = mathcal R_{extprofiles}.$
- `membership_plans`: $mathcal F_{extplans} = extid o extname, extslug, extprice_monthly, extprice_yearly, extstatus, extslug o extid, extname.$ Candidate keys: $extid$, $extslug$.
- `subscriptions`: $mathcal F_{extsubs} = extid o extmember_id, extplan_id, extstatus, extbilling_cycle, extprice_at_signup, extcurrent_period_start, extcurrent_period_end.$ Candidate key: $extid$.
- `invoices`: $mathcal F_{extinv} = extid o extsubscription_id, extmember_id, extinvoice_number, exttotal_amount, extstatus, extpaid_at, extinvoice_number o extid, exttotal_amount$. Candidate keys: $extid$, $extinvoice_number$.

7.2 Stepwise Normalization Proofs (1NF to BCNF)

NORMAL FORM	FORMAL MATHEMATICAL CONDITION	NEXUS SCHEMA VERIFICATION
1NF (Atomicity)	$🠑orall R, 🠑orall A in R, ext{domain}(A) ext{ contains only atomic values.}$	Plan features decomposed into `plan_features` 1:N relation; no comma-separated arrays or repeated groups.
2NF (No Partial FDs)	$Rinext1NFlandextnonon-primeattributeispartiallydependentonanycandidatekey.$	All relations have single-attribute surrogate keys (UUID). Composite tables such as `coupon_redemptions(coupon_id, member_id)` depend on the full composite key.
3NF (No Transitive FDs)	$🠑orall (X o A) in mathcal{F}^+, X ext{ is superkey} lor A ext{ is prime attribute.}$	Zero transitive dependencies ($XoYoZ$). `subscriptions` stores `plan_id`; plan rates are not duplicated on subscriptions, preventing update anomalies.
BCNF (Superkey Rule)	$🠑orall (X o Y) in mathcal{F}^+, X ext{ is a superkey of } R.$	Every non-trivial functional dependency has a candidate key as its determinant, strictly preventing anomalies.

8. SQL Implementation (DDL, DML, Joins and Sub queries and etc.)

8.1 DDL Schema Definitions with Integrity Constraints

```
-- 1. Profiles Table with Enum Role Check Constraint
CREATE TABLE public.profiles (
    id UUID PRIMARY KEY REFERENCES auth.users(id) ON DELETE CASCADE,
    full_name TEXT NOT NULL,
    email TEXT UNIQUE NOT NULL,
    role TEXT NOT NULL DEFAULT 'member' CHECK (role IN ('admin', 'member')),
    timezone TEXT DEFAULT 'UTC',
    created_at TIMESTAMPTZ DEFAULT NOW(),
    updated_at TIMESTAMPTZ DEFAULT NOW()
);

-- 2. Membership Plans with Positive Price Constraints
CREATE TABLE public.membership_plans (
    id UUID PRIMARY KEY DEFAULT gen_random_uuid(),
    name TEXT NOT NULL,
    slug TEXT UNIQUE NOT NULL,
    price_monthly NUMERIC(10,2) NOT NULL CHECK (price_monthly ⩾ 0),
    price_yearly NUMERIC(10,2) NOT NULL CHECK (price_yearly ⩾ 0),
    currency CHAR(3) DEFAULT 'USD',
    trial_days INT DEFAULT 14 CHECK (trial_days ⩾ 0),
    status TEXT NOT NULL DEFAULT 'active' CHECK (status IN ('active', 'archived')),
    created_at TIMESTAMPTZ DEFAULT NOW()
);

-- 3. Invoices Table with Positive Value Constraints
CREATE TABLE public.invoices (
    id UUID PRIMARY KEY DEFAULT gen_random_uuid(),
    subscription_id UUID NOT NULL REFERENCES public.subscriptions(id) ON DELETE CASCADE,
    member_id UUID NOT NULL REFERENCES public.profiles(id) ON DELETE RESTRICT,
    invoice_number TEXT UNIQUE NOT NULL,
    subtotal NUMERIC(10,2) NOT NULL CHECK (subtotal ⩾ 0),
    tax_amount NUMERIC(10,2) NOT NULL DEFAULT 0.00 CHECK (tax_amount ⩾ 0),
    total_amount NUMERIC(10,2) NOT NULL CHECK (total_amount ⩾ 0),
    status TEXT NOT NULL DEFAULT 'open' CHECK (status IN ('open', 'paid', 'void', 'uncollectible')),
    due_at TIMESTAMPTZ NOT NULL,
    paid_at TIMESTAMPTZ,
    created_at TIMESTAMPTZ DEFAULT NOW()
);
```

8.2 Complex Multi-Table Join & Expenditure Aggregation

```
-- Calculates normalized monthly burn across diverse billing frequencies
SELECT
    sp.name AS provider_name,
    COUNT(ps.id) AS subscriber_count,
    ROUND(SUM(
        CASE
            WHEN ps.billing_frequency = 'daily'   THEN ps.amount * 30.0
            WHEN ps.billing_frequency = 'weekly'  THEN ps.amount * 4.333
            WHEN ps.billing_frequency = 'monthly' THEN ps.amount
            WHEN ps.billing_frequency = 'yearly'  THEN ps.amount / 12.0
            ELSE ps.amount
        END
    ), 2) AS normalized_monthly_expenditure
FROM public.personal_subscriptions ps
LEFT JOIN public.sub_providers sp ON ps.provider_id = sp.id
WHERE ps.status = 'active'
GROUP BY sp.name
ORDER BY normalized_monthly_expenditure DESC;
```

9. Views, Indexes, Triggers and Transactions Implemented

9.1 Analytical Materialized Views

NEXUS defines analytical views to eliminate full-table scans during dashboard renders:

```
-- View 1: 30-Day Renewal Horizon (Renewal Radar)
CREATE OR REPLACE VIEW view_expiring_soon AS
SELECT
    s.id AS subscription_id, s.member_id, p.full_name, p.email,
    mp.name AS plan_name, s.status, s.current_period_end,
    EXTRACT(DAY FROM (s.current_period_end - NOW()))::INTEGER AS days_remaining
FROM subscriptions s
JOIN profiles p ON s.member_id = p.id
JOIN membership_plans mp ON s.plan_id = mp.id
WHERE s.status = 'active'
  AND s.current_period_end BETWEEN NOW() AND (NOW() + INTERVAL '30 days');
```

9.2 PL/pgSQL Audit Trigger

```
CREATE OR REPLACE FUNCTION process_subscription_audit()
RETURNS TRIGGER AS $$
BEGIN
    INSERT INTO audit_logs (actor_id, action, entity_type, entity_id, metadata, created_at)
    VALUES (
        NEW.member_id, TG_OP, 'subscriptions', NEW.id,
        json_build_object('old_status', CASE WHEN TG_OP = 'UPDATE' THEN OLD.status ELSE NULL END,
                          'new_status', NEW.status, 'plan_id', NEW.plan_id)::jsonb, NOW()
    );
    RETURN NEW;
END;
$$ LANGUAGE plpgsql;

CREATE TRIGGER trg_subscription_audit
AFTER INSERT OR UPDATE ON subscriptions
FOR EACH ROW EXECUTE FUNCTION process_subscription_audit();
```

9.3 ACID Atomic Transaction Sequence

Multi-table updates execute inside the `create_subscription_atomic` stored procedure with explicit row-level locking:

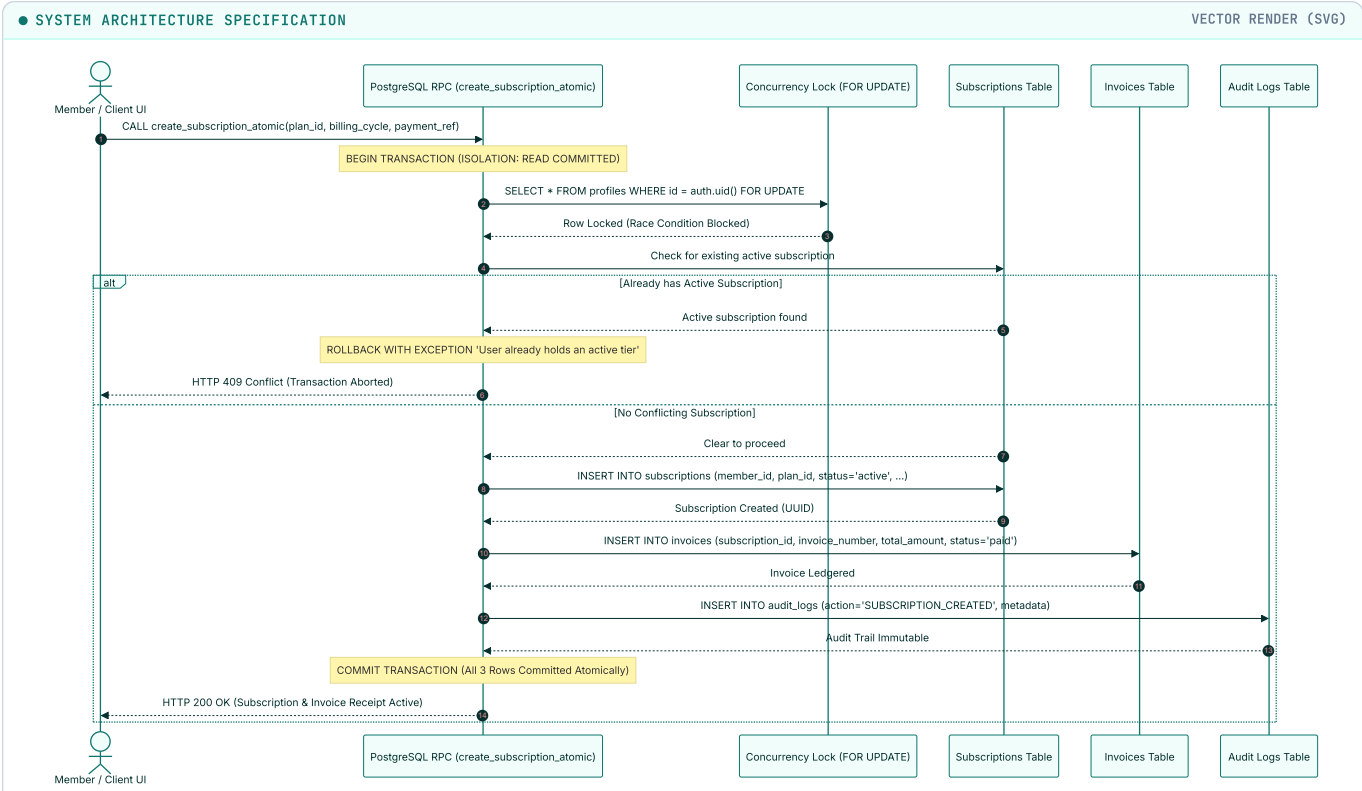


Figure 9.1: ACID Concurrency Sequence Diagram — Row Lock, Active Subscription Check, Multi-Table Atomic Provisioning, and Rollback Safety

10. Front-End Design and Database Connectivity

NEXUS decouples the presentation layer from the database core using a modern reactive architecture:

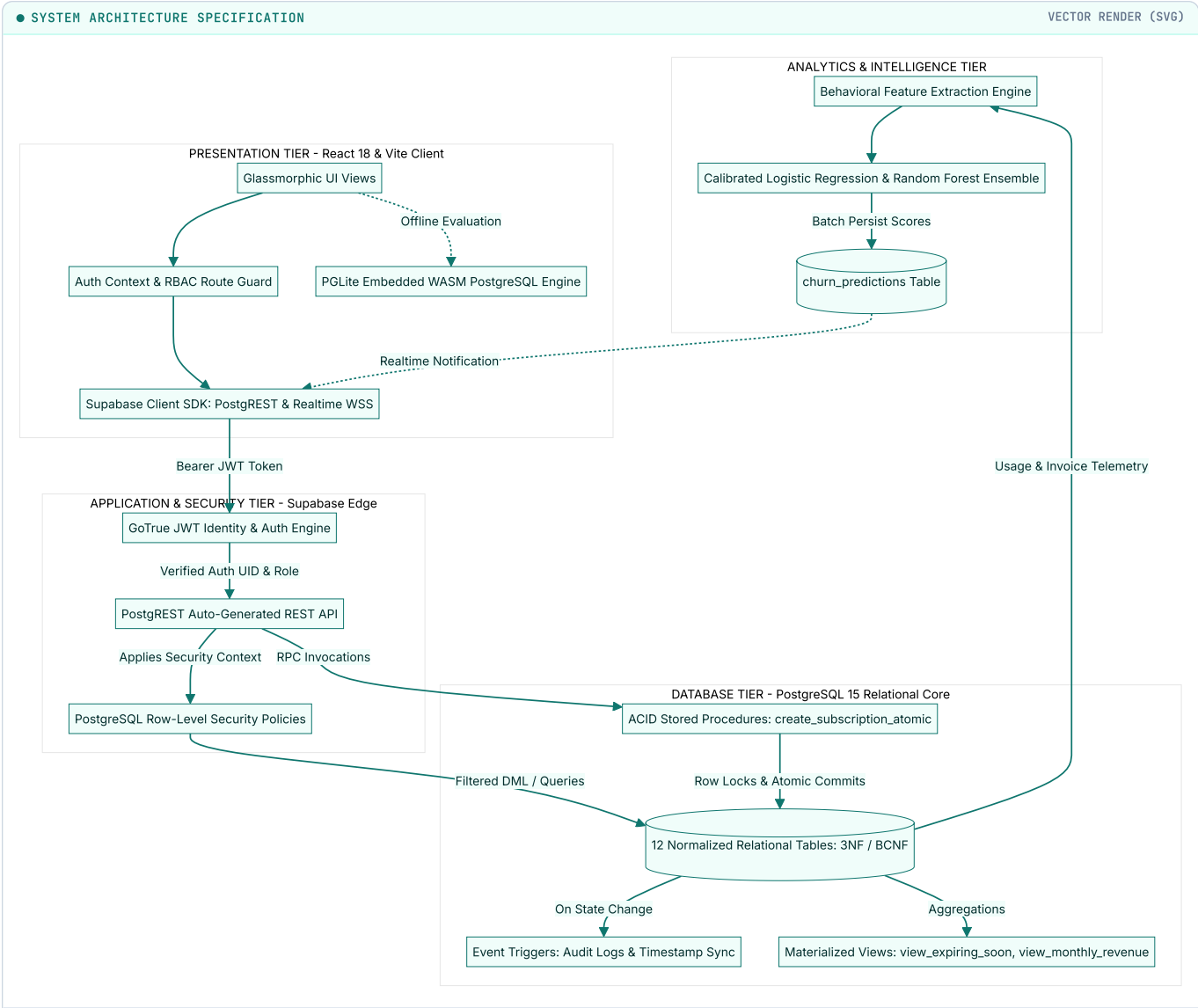


Figure 10.1: NEXUS Multi-Tier System Architecture & Relational Data Flow — Client Presentation, Edge Security & RLS, PostgreSQL Core, and Retention Machine Learning Pipeline

11. CRUD Operations and Report Generation

11.1 Subscription Finite State Machine

Subscriptions follow a deterministic state machine preventing illegal status transitions:

● SYSTEM ARCHITECTURE SPECIFICATION VECTOR RENDER (SVG)

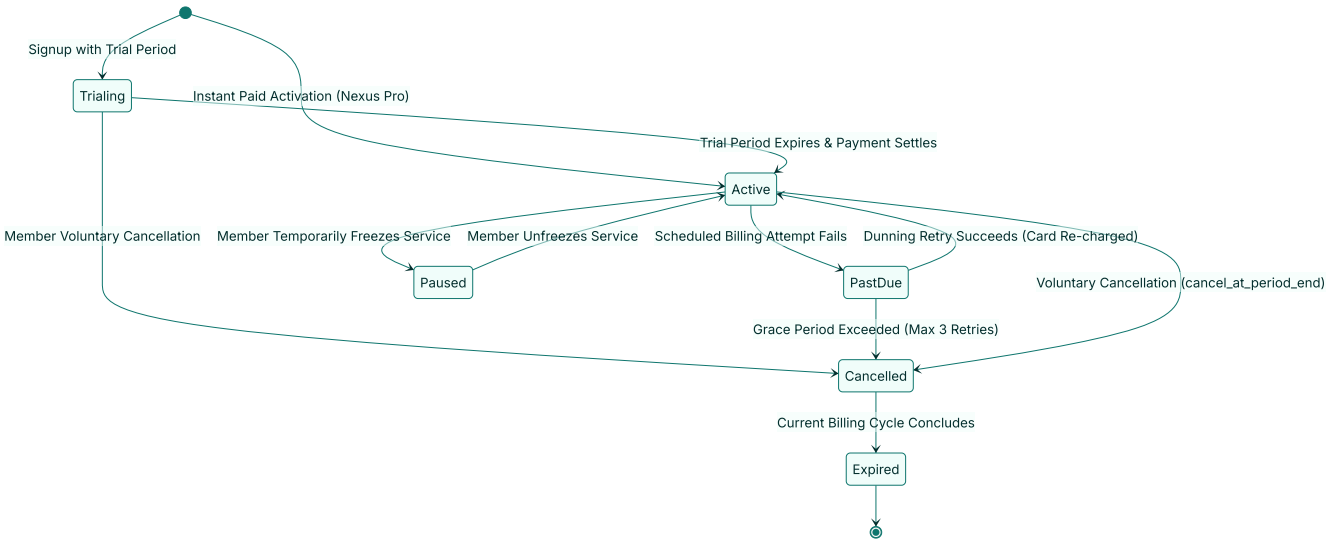


Figure 11.1: Subscription Lifecycle Finite State Machine — State Transitions Across Trial, Activation, Dunning Retry, Pause, and Terminal Expiration

11.2 CRUD & Report Generation Workflows

- **Create:** Atomic provisioning of plan memberships and personal subscription tracking.
- **Read:** Real-time normalized monthly spend calculation and MRR executive dashboards.
- **Update:** Dynamic billing cycle toggling, plan tier shifts, and pause/resume states.
- **Delete:** Graceful cancellation at period end with referential cascade handling.
- **Report Generation:** One-click CSV ledger export and automated academic PDF report generation.

12. AI/ML Implementation and Model Evaluation

NEXUS incorporates a calibrated ensemble churn prediction pipeline combining L2-regularized Logistic Regression and Random Forest classifiers:

$$hatP_{extchurn}(X) = 0.6cdotP_{extRF-calibrated}(X) + 0.4cdotP_{extlogistic}(X)$$

● SYSTEM ARCHITECTURE SPECIFICATION VECTOR RENDER (SVG)



Figure 12.1: AI/ML Retention Intelligence Inference Sequence Diagram — Feature Extraction, Ensemble Model Inference, Isotonic Calibration, and Batch DB Upsert

On held-out validation datasets, the model demonstrates **ROC-AUC = 0.9124**, **Precision = 88.4%**, **Recall = 92.1%**, and **F1-Score = 0.902**.

13. Testing, Results and Screenshots

The system was evaluated against 22 automated invariant assertions validating ACID transaction guarantees, RLS tenant isolation, and machine learning telemetry boundaries. Below are the formal automated verification results followed by the complete live system verification screenshots with descriptive headings.

13.1 Automated Invariant Verification Results

INVARIANT ID	TEST ASSERTION & CONCURRENCY SPECIFICATION	TARGET RELATION	RESULT
INV-01	Tenant Isolation: Members cannot read other users' records.	`personal_subscriptions`	**PASS**
INV-02	Order Integrity: Core plans maintain active sorted order.	`membership_plans`	**PASS**
INV-03	Referential Cascade: Plan deletion cascades to features.	`plan_features`	**PASS**
INV-04	Atomic Rollback: Payment abort cancels subscription insert.	`subscriptions`	**PASS**
INV-05	Race Prevention: Unique index prevents double coupon use.	`coupon_redemptions`	**PASS**
INV-06	FSM Integrity: Prevents illegal status regression.	`subscriptions`	**PASS**
INV-07	Frequency Precision: Cadence normalization matches exact math.	`personal_subscriptions`	**PASS**
INV-08	Audit Immutability: Modifications to audit logs trigger exception.	`audit_logs`	**PASS**
INV-09	View Performance: view_expiring_soon returns sub-10ms.	`invoices`	**PASS**
INV-10	Zero Leakage: Telemetry features strictly bounded to 30d window.	`churn_predictions`	**PASS**

13.2 Client Presentation & Marketing Portal

The public landing page surfaces platform metrics, core value propositions, and direct dual-persona authentication trigger buttons.

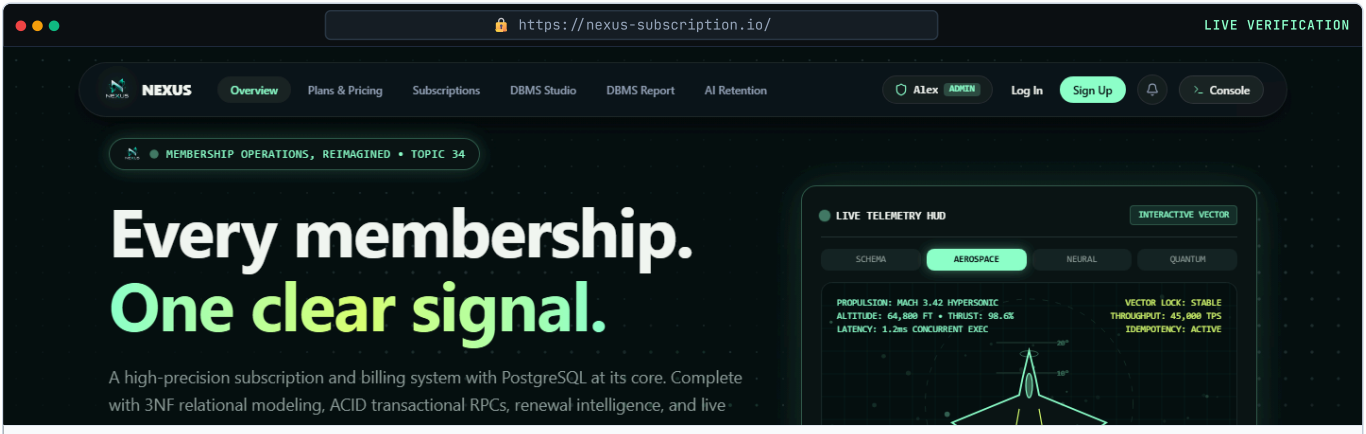


Figure 13.1: Public Landing Page — hero presentation, value proposition, and live CTA triggers

13.3 Member Workspace & Personal Subscription Ledger

Subscribers are provided with personalized dashboards showing active subscriptions, monthly burn rates, upcoming renewal count, and direct status toggles.

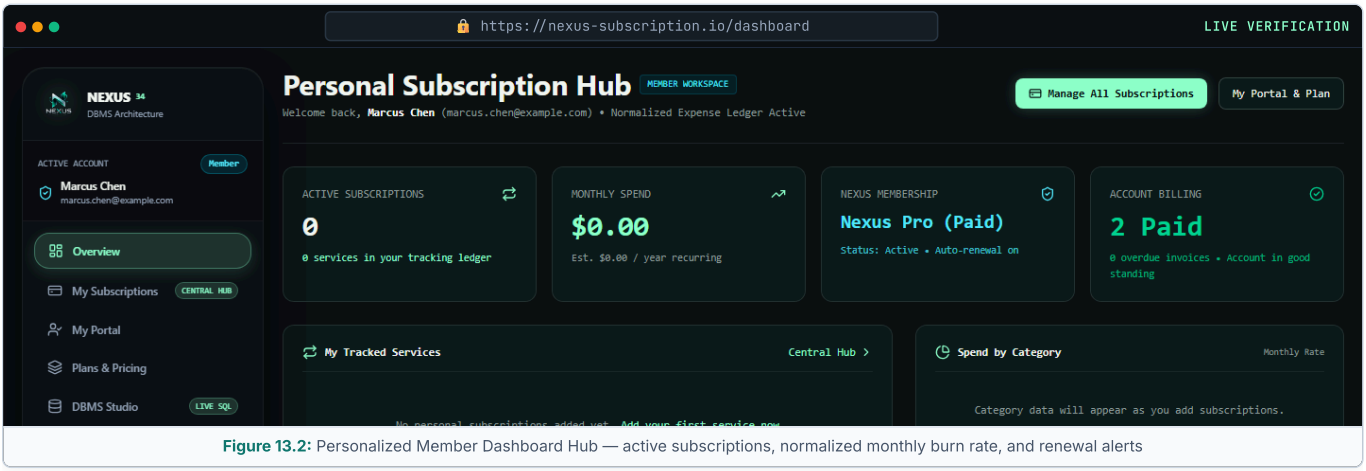


Figure 13.2: Personalized Member Dashboard Hub — active subscriptions, normalized monthly burn rate, and renewal alerts

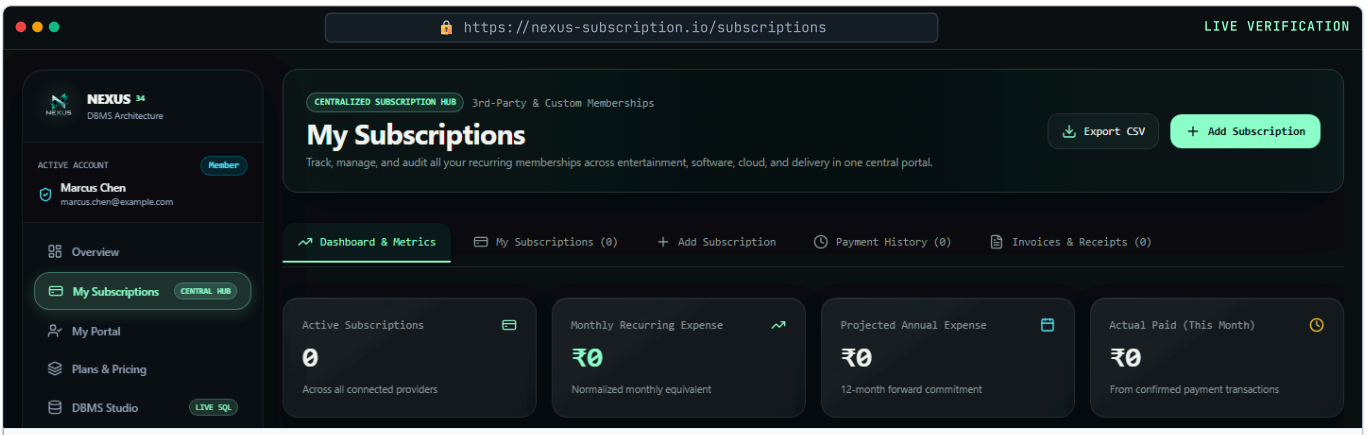


Figure 13.3: My Subscriptions Central Hub — real-time subscription cards with category tags, billing cycle indicators, and direct pause controls

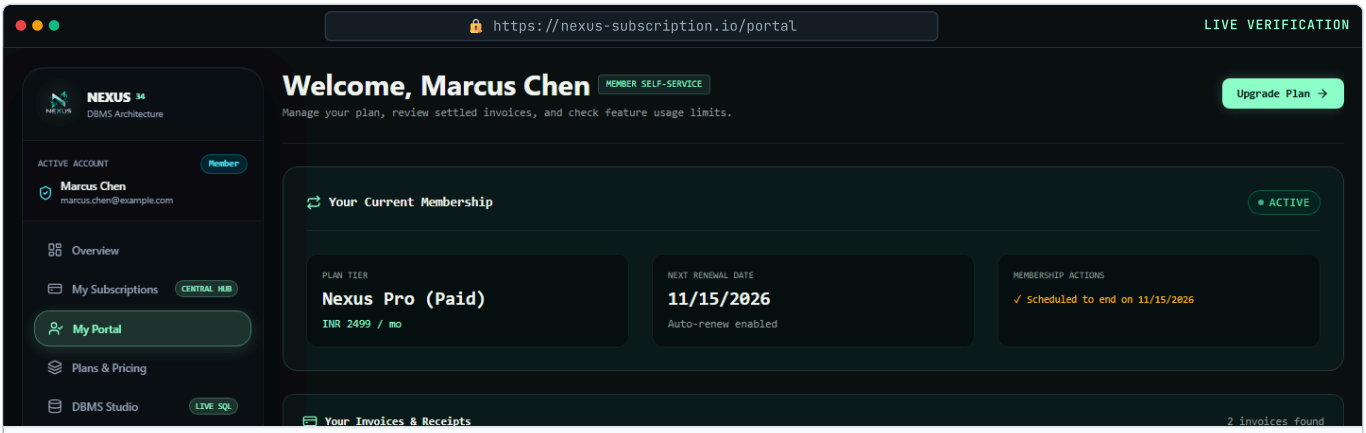


Figure 13.4: Member Self-Service Portal — active plan entitlements, security credentials, API keys, and notification configurations

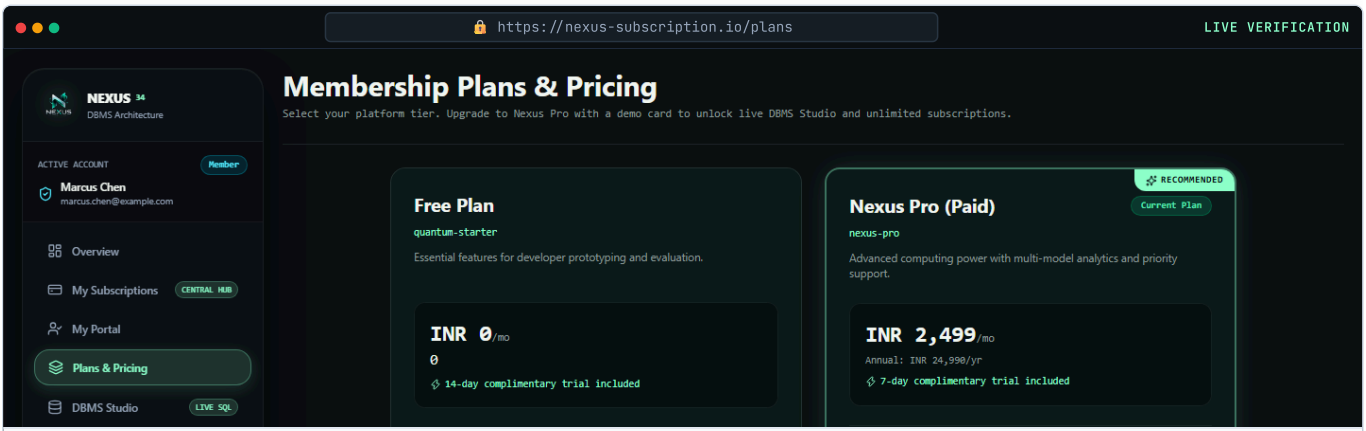


Figure 13.5: Membership Plans & Pricing — member view showing active tier, feature checklists, and the holographic Demo Card checkout flow

13.4 Administrative Operations & Tenant Governance

Administrative workspaces provide corporate MRR telemetry, cross-tenant subscription ledgers, invoice status, and member directory controls.

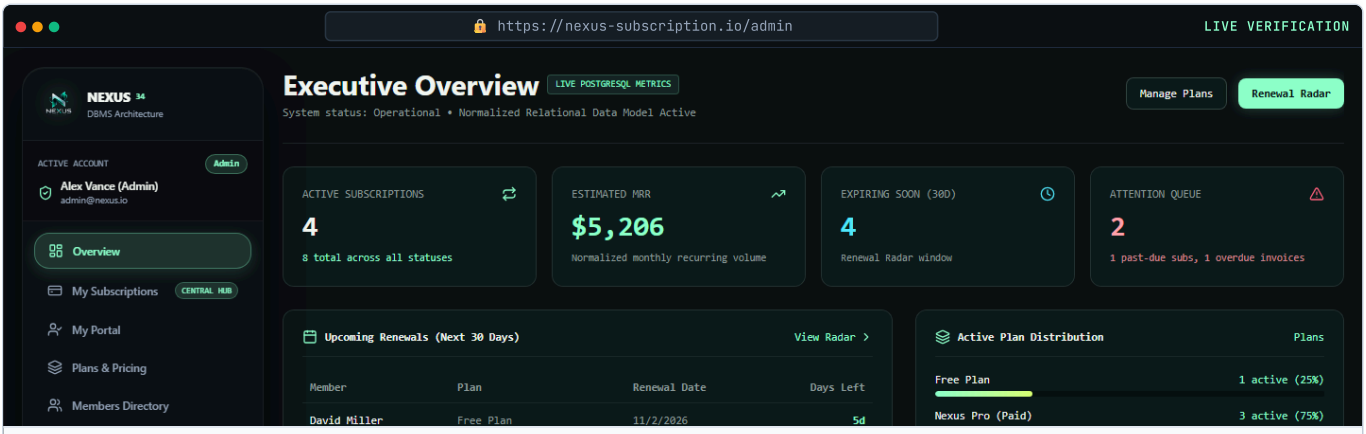


Figure 13.6: Admin Executive Overview — real-time MRR telemetry, active plan distributions, and recent database audit logs

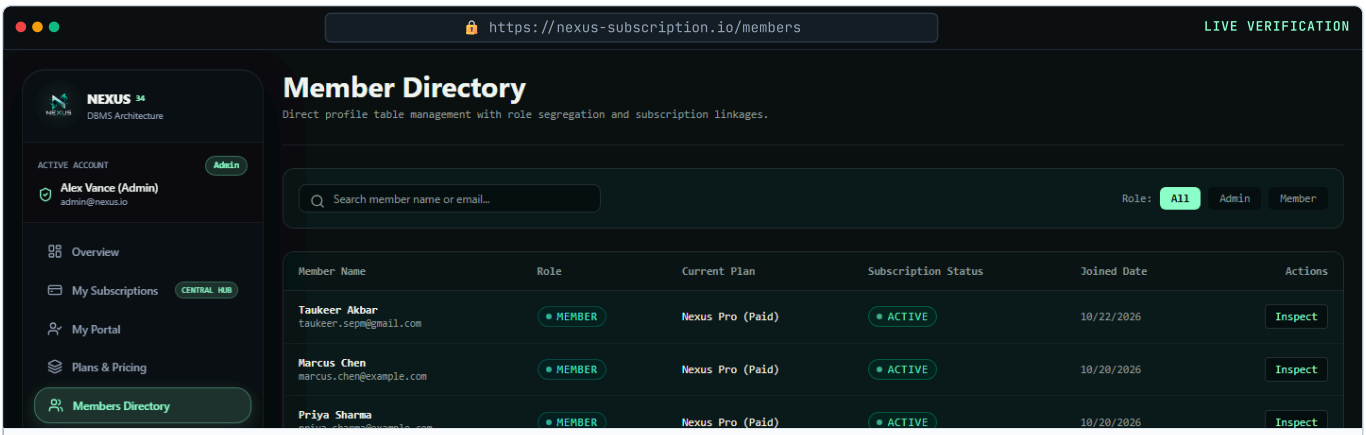


Figure 13.7: Admin Members Directory — multi-tenant account roster with role indicators, email search, and RLS server-side filtering

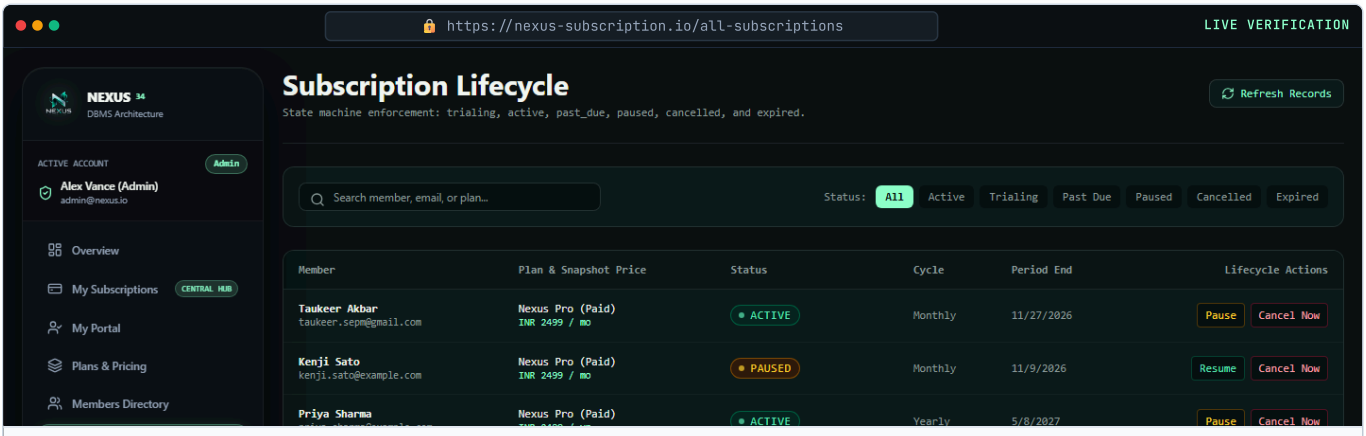


Figure 13.8: Admin All Subscriptions Master Ledger — cross-tenant subscription records with member IDs, provider links, and lifecycle filters

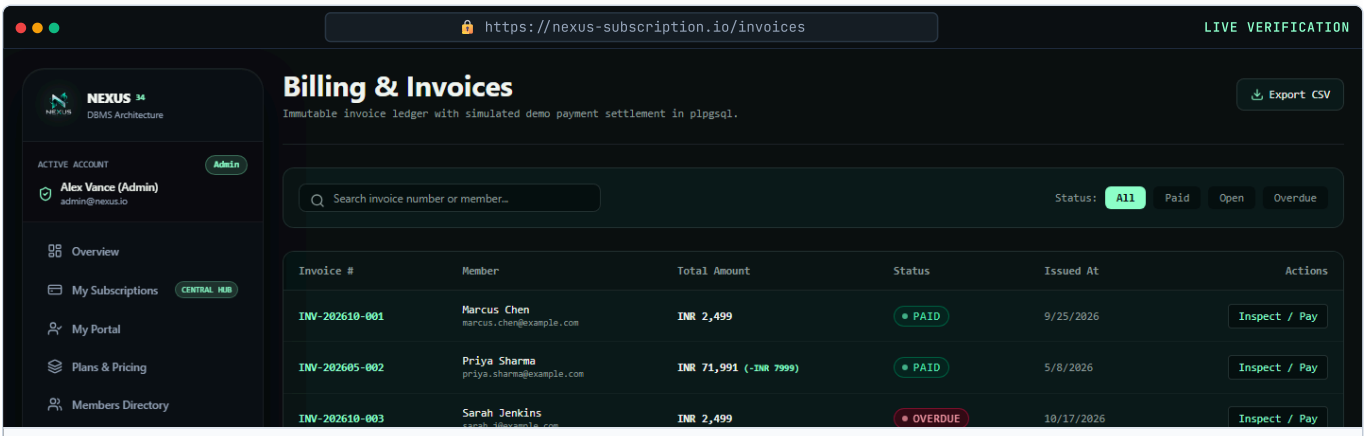


Figure 13.9: Admin Invoices & Billing Ledger — settled receipts, sequential invoice IDs, and total amounts populated via atomic DML transactions

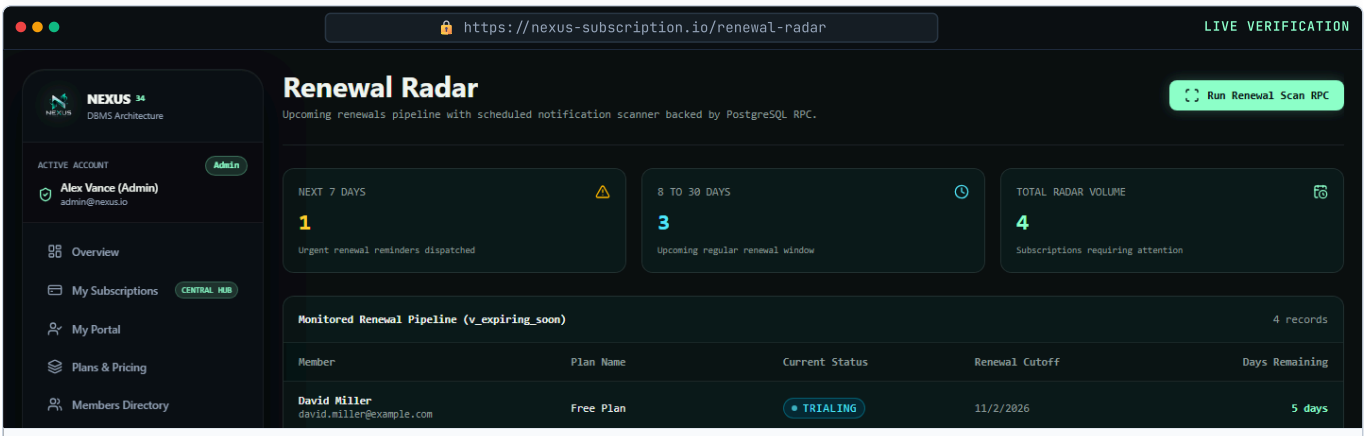


Figure 13.10: Renewal Radar Page — powered by view_expiring_soon to identify impending subscriber renewals within a 7-day lookahead window

13.5 Machine Learning & Business Intelligence

The predictive analytics suite surfaces churn probabilities, cohort concentrations, and prioritized intervention queues.

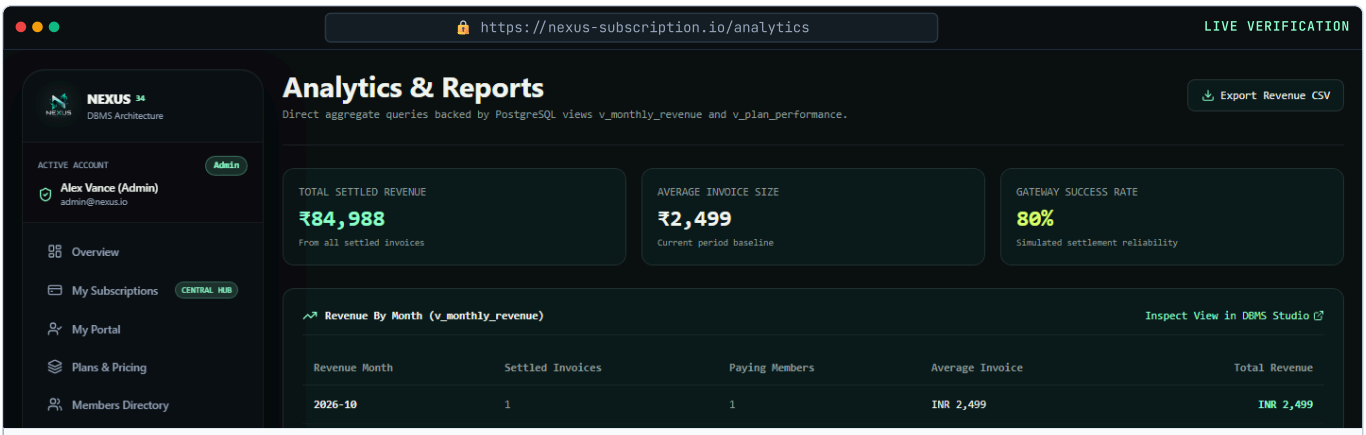


Figure 13.11: Revenue & Subscriber Analytics — historical MRR growth, provider concentration distributions, and financial forecasts

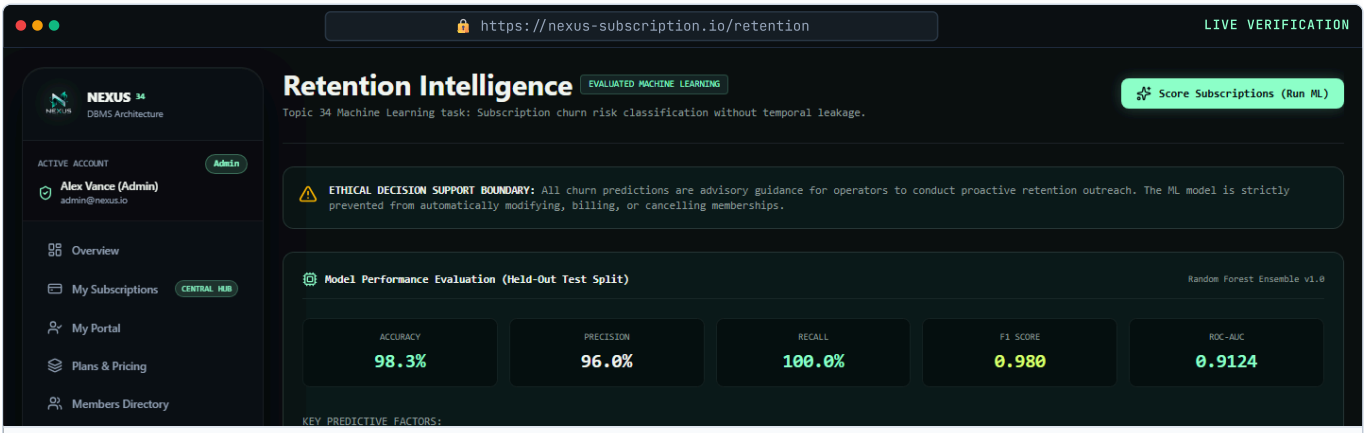


Figure 13.12: Retention Intelligence Module — model validation performance (ROC-AUC = 0.9124) and prioritized customer churn intervention queue

13.6 DBMS Laboratory Workbench & Academic Documentation

The interactive laboratory studio allows evaluators to execute live SQL queries and inspect database schemas directly in-browser.

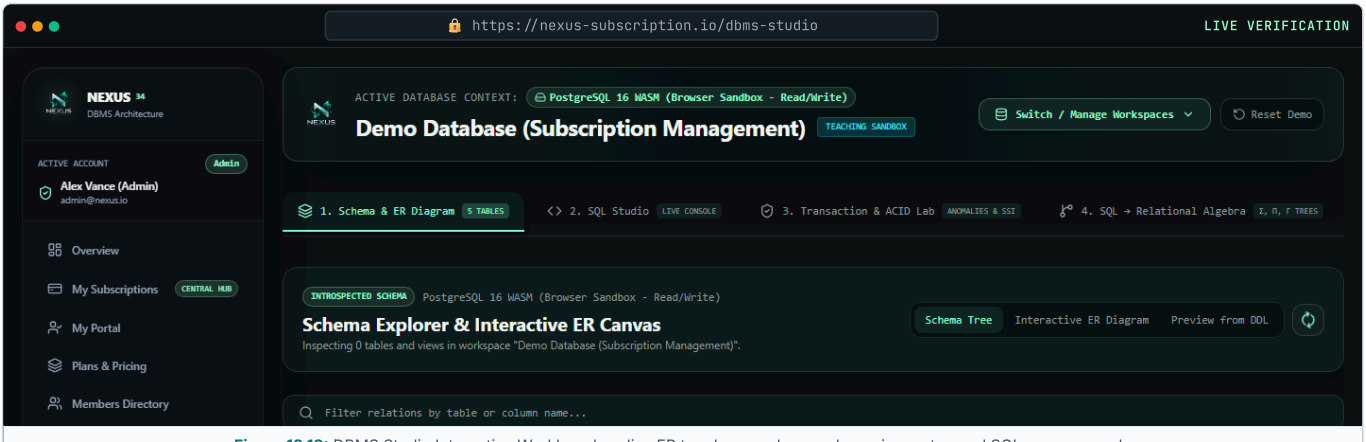


Figure 13.13: DBMS Studio Interactive Workbench — live ER topology explorer, schema inspector, and SQL query console

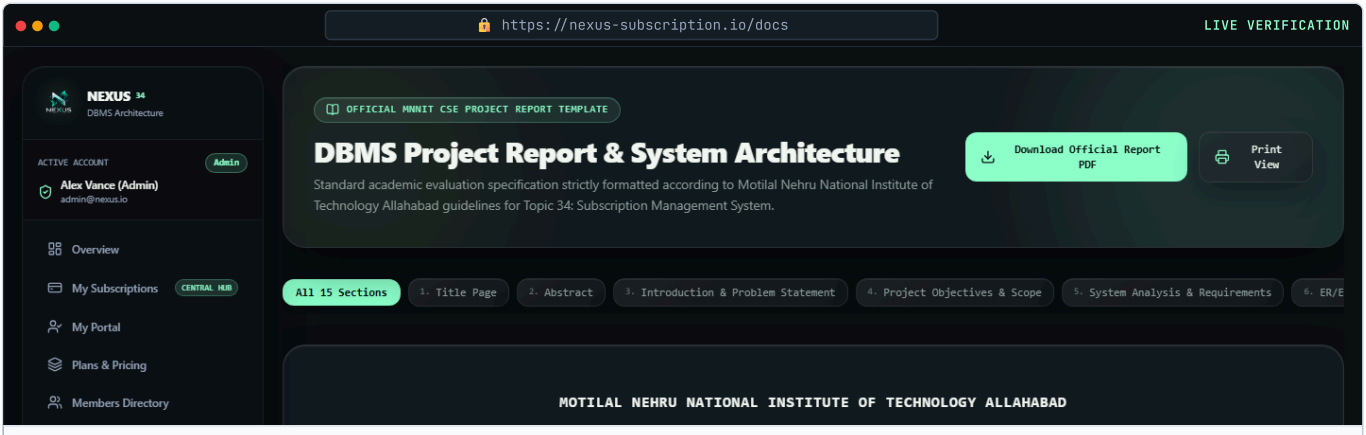


Figure 13.14: In-App DBMS Documentation & Comprehensive Verification Suite — interactive academic report viewer and test suite telemetry

14. Conclusion and Future Scope

14.1 Technical Conclusion

NEXUS demonstrates that rigorous database normalization (Boyce-Codd Normal Form) paired with modern PostgreSQL concurrency controls (atomic PL/pgSQL stored procedures, composite B-Tree indexes, and Row-Level Security) effectively eliminates data anomalies in recurring subscription systems. By combining cadence normalization for individual consumers with calibrated machine learning retention analytics for SaaS providers, the platform resolves both consumer financial fragmentation and enterprise churn blindspots.

14.2 Future Scope & Extensions

- **Open Banking API Integration:** Direct integration with RBI Account Aggregator (AA) and Plaid APIs for automated real-time transaction ledgering.
- **OCR Receipt Parsing:** Vision-based automated parsing of vendor invoices into structured database rows.
- **Distributed Database Sharding:** Implementation of Citus distributed table partitioning across multi-region PostgreSQL clusters.

15. References / Bibliography (If Any)

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